# Hima Shanthi Rayapati

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## Education

**Arizona State University**, Data Science, Analytics and Engineering (Human-Centered Applications) 08/2026 – 12/2027

- Fulton Schools of Engineering Graduate Scholarship | ASU AI Scholars Program, 2026–27

**Arizona State University**, BS in Psychology and Mathematics (Statistics) 01/2023 – 05/2026

- **Minor:** Philosophy | **GPA:** 4.0/4.0

## Research Experience

**Research Assistant**, *CARE Lab (Prof. Qiaoning Zhang)*, ASU 08/2026 – Present

- Attend lab meetings and discuss ongoing research on human-AI interaction.
- Building a multimodal AI tutor where students choose how they learn (visual, text, or speech), to use in a study comparing it with a standard AI tutoring chatbot (in progress).

**UX Researcher (UXR Guild)**, *NEXT Lab*, ASU 08/2026 – 10/2026

- Tested games, digital twins, and interactive apps with users to find usability problems and collect feedback.

**Research Assistant**, *Thinking Across Languages and Contexts Lab*, ASU 08/2025 – 10/2026

- Assisted a graduate student with testing experiments built in jsPsych and JavaScript.
- Helped check whether data were recorded correctly and organized for analysis.
- Performed exploratory data analysis and assisted in interpreting experimental results under supervision.

**Research Assistant**, *Perception, Action, Language & Meaning Lab*, ASU 08/2025 – 05/2026

- Ran two-person collaborative task sessions (e.g., Taboo), observing and recording coordination, communication, and performance data using VR, motion-capture, and eye-tracking.
- Worked with the lab's time-series behavioral data, learning how to clean, organize, and explore it.
- Read papers and explored possible methods (e.g., multilevel vector autoregression) to understand which approaches might fit the lab's data.
- Contributed to discussions by sharing preliminary notes and observations from trial analyses.

**Volunteer Research Assistant**, *Longitudinal & Behavioral Data Science Lab*, ASU 08/2024 – 12/2024

- Participated in bi-weekly lab meetings, contributing to discussions and staying updated on research progress.

**Research Assistant**, *Evolution, Ecology, and Social Behavior Lab*, ASU 08/2023 – 11/2023

- Assisted with survey design, pilot testing, data coding, and literature reviews in social cognition research.
- Participated in lab meetings and contributed to discussions on study design and interpretation.

## Projects

**Trust in Advice: AI vs. Human Sources (Data Analysis & Written Report)** ***Trust in AI.com/2025***

**Tools:** SAS (PROC GLMSELECT, PROC GLM) · **Methods:** General linear model, Stepwise selection (AIC), Residual diagnostics

- Analyzed 30,000+ human-AI interactions to study effects of advice source.
- Modeled decision changes before and after advice exposure.
- Found advice correctness as primary driver, with modest AI-related effects.

**Carryover: AI-Supported Daily Living App for Autistic Adults** ***GitHub***

**Tools:** Python, React, FastAPI, pytest · **Methods:** User research, Task analysis, Literature review, Rule-based prompt fading

- Led user research with partner First Place AZ (a site visit and 2 staff meetings), and turned the findings into design changes, including always-visible safety cautions and adaptive help per step.
- Wrote a literature review on prompt fading and executive function, and wrote the fading rules (when help fades or comes back, per step and per person) that the app follows.
- Built a Python module that detects where users get stuck (4 patterns) from interaction logs, requiring patterns to repeat before acting, to avoid false alarms.
- Co-built a working prototype with simulated data (React, FastAPI, an LLM with code-based safety rules, and automated tests) on a 5-person team, and presented it at the Scottsdale Launchpad.

**Detecting and Correcting Social Bias in AI Question Answering (CSE 575, in progress)**

**Methods:** Text classification, Bias evaluation, Data preprocessing

- Team project building a pipeline that detects biased answers from LLMs on the BBQ bias benchmark and rewrites them, while keeping accuracy on questions the context can answer.
- Responsible for the data side: auditing and preprocessing BBQ, a loader that converts it into a general (text, label) format, and template-grouped train/validation/test splits so near-duplicate questions don't leak into the test set.

**Cultural Differences in Trust: AI vs. Human Expert Decision-Making** ***Research Proposal / 2025***

**Methods:** Mixed-effects modeling, Experimental design, Pre-registration

- Proposed cross-cultural study (Mexico vs. U.S.) on AI vs. human decision-making under conflict.
- Outlined experimental design focusing on competence and legitimacy.
- Formulated mixed-effects modeling approach for analysis.
- Synthesized literature across cognition, decision-making, and human-AI interaction.

**Dual-Timescale Model of Learning (ODE & SDE)** ***GitHub***

**Tools:** MATLAB, R (Shiny, deSolve, ggplot2) · **Methods:** ODE/SDE modeling, Stability analysis, Euler-Maruyama simulation

- Part 1 (solo): built a model of fast and slow learning, analyzed its stability, and simulated it in MATLAB.
- Part 2 (with a partner): did the theory for a stochastic version of the model (SDEs, Euler-Maruyama).
- Built R Shiny apps to explore the model, including a 3-tab app for Part 2 that is live online.

**Probabilistic Reward Learning Task using PsychoPy** ***Reward Learning.com/2025***

**Tools:** PsychoPy, Python (pandas, NumPy) · **Methods:** Experimental design, Learning under uncertainty

- Designed probabilistic reward-learning task with informative vs. non-informative features.
- Examined learning under uncertainty and perceptual noise.
- Collected and analyzed participant data using Python.

*Additional projects and full write-ups available at* my portfolio.

## Work Experience

**Graduate Course Assistant (Statistical Methods)**, ASU – Tempe, AZ 10/2026 – Present

- One of two course assistants for PSY 330 – Statistical Methods.
- Grade assignments and exams with written feedback, following the instructor's grading guidelines.
- Answer student emails about course content and SPSS.

**Research Aide**, ASU – Tempe, AZ 12/2024 – Present

- Review, categorize, and structure user-generated data to ensure quality and consistency.
- Assign relevant tags to comments for improved organization and searchability.
- Check whether AI-generated tags follow the evaluation guidelines and are accurate.
- Perform data cleaning tasks, identifying and correcting inconsistencies to maintain data accuracy.
- Work with databases to structure data and ensure accessibility for future use.

## Skills

- **Programming & Statistical Tools:** R, SAS, MATLAB, Python (pandas, NumPy, statsmodels, scikit-learn, Matplotlib), JMP, JASP, SQL (Microsoft Access), Java, C, Bash, REST APIs
- **Machine Learning, NLP & AI Evaluation:** scikit-learn, Classification (logistic regression, random forest, SVM), Ridge regression, Clustering, Dimensionality reduction (PCA), Neural networks (fundamentals), Feature engineering, Hyperparameter tuning, Cross-validation, Model evaluation (precision/recall/F1, ROC-AUC), Text classification, Basic NLP, LLM output evaluation, Bias and fairness evaluation, Anomaly detection, Time-series analysis
- **Experimental & Research Tools:** jsPsych, PsychoPy, JavaScript, HTML, CSS, Qualtrics, SurveyMonkey
- **Data Analysis & Visualization:** Tableau, Excel, Google Sheets, Alteryx
- **Software & Development:** React, FastAPI, pytest, Git/GitHub
- **Design & Prototyping:** Figma, Figma Motion